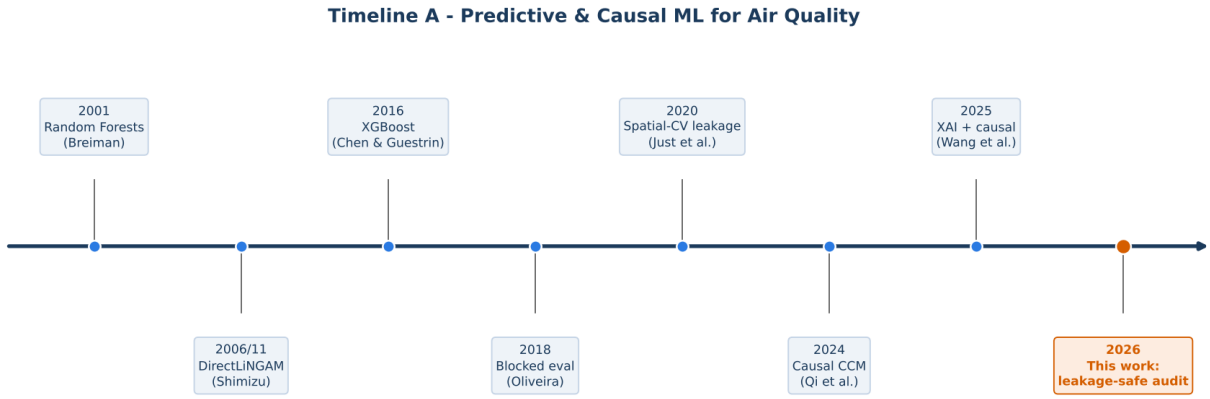
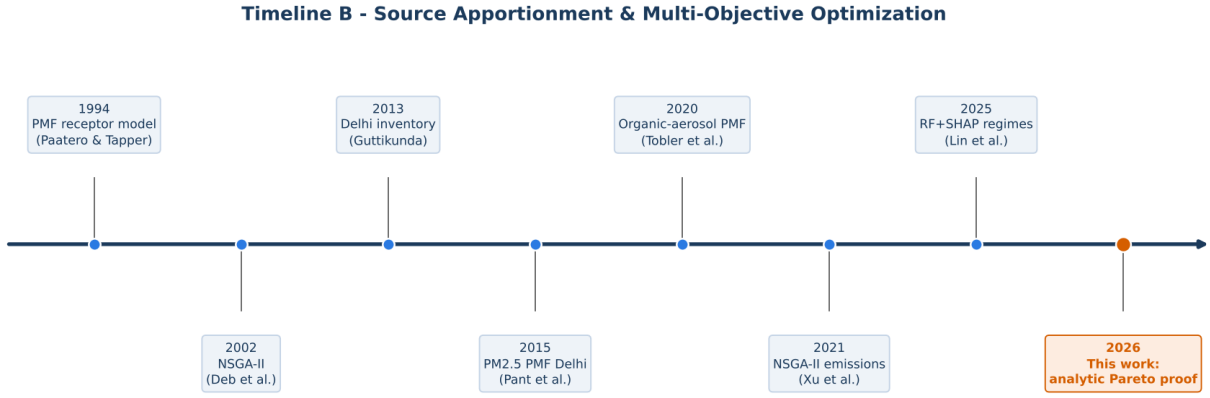
MDP Project

Literature Review & Research Gaps

Two timelines • gaps we solve • contributions & novelty • SCI/Q1 reading list



Timeline A - predictive & causal ML for air quality.



Timeline B - source apportionment & multi-objective optimization.

Literature Review, Timelines, Gaps, Contributions & Novelty

Prepared against the Review-2 checklist: *introduction gaps, research paper on a timeline (two timelines), the gaps we are solving, and our contributions & novelty*, grounded in SCI / Q1 literature.

> ***How to read the "two timelines":*** *the field our paper sits in has two parallel method threads.* ***Timeline A*** *traces *predictive & causal machine learning for air quality*;* ***Timeline B*** *traces *source apportionment & multi-objective policy optimization*. Our work sits at the end of both and connects them. Each timeline below lists, per milestone, the contribution and the limitation it leaves open; those open limitations are synthesized into the research gaps in §3.*

---

1. Timeline A -- Predictive & Causal ML for Air Quality

Year	Milestone (author, venue)	Contribution	Limitation it leaves
2001	Breiman, *Machine Learning*	Random forests: strong non-linear tabular prediction	Associational; no mechanism
2006/11	Shimizu et al., *JMLR* (LiNGAM / DirectLiNGAM)	Orient causal edges from non-Gaussianity	Assumes no hidden confounders, linearity
2016	Chen & Guestrin, *KDD* (XGBoost)	De-facto standard for tabular AQ prediction	Accuracy prioritised over interpretability
2017	Lundberg & Lee, *NeurIPS* (SHAP)	Principled feature attributions	Explains a model, not a mechanism
2018	Oliveira et al., *ECML-PKDD*	Dependence-aware (blocked) evaluation for spatio-temporal data	Not yet standard practice in AQ-ML
2019	Grange & Carslaw, *Sci. Total Environ.*	Random-forest meteorological normalisation	Meteorology must be removed before attribution
2020	Just et al., *Remote Sensing*	Shows random CV inflates PM2.5 skill vs spatial CV (RMSE 2.10 vs 3.11)	Leakage still widespread in practice
2024	Qi et al., *Environ. Res. Lett.*	Convergent cross-mapping for PM2.5 <-> O3 causality (China)	City/season-specific; not identification
2024	Rautela & Goyal, *Sci. Reports*; Malhotra et al., *J. Big Data*	Reviews of AI/ML for AQ (incl. India)	Confirm accuracy-first framing; few leakage-safe
2025	Wang et al., *Environ. Sci. Ecotechnol.*	XAI + causal inference for the 2022 ozone rebound	Correlational drivers, not interventions
2026	Ye et al., *Atmos. Chem. Phys.*	SHAP ozone-trend attribution; meteorology dominant	Flags SHAP instability under multicollinearity
2026	**This work**	**Leakage-safe India benchmark + causal-stability audit + identification gate**	--

2. Timeline B -- Source Apportionment & Multi-Objective Policy Optimization

Year	Milestone (author, venue)	Contribution	Limitation it leaves
1994	Paatero & Tapper, *Environmetrics* (PMF)	Receptor modelling with error weighting	Needs speciation + per-measurement uncertainties
1999	Zitzler & Thiele, *IEEE TEC*	Hypervolume for Pareto quality	Metric, not a guarantee of optimality
1999	Sillman, *Atmos. Environ.*	VOC- vs NOx-limited ozone regimes	Requires VOC observations to resolve
2002	Deb et al., *IEEE TEC* (NSGA-II)	Standard multi-objective evolutionary search	No guarantee vs exact fronts on simple surfaces
2013	Guttikunda & Calori, *Atmos. Environ.*	1 km Delhi emission inventory	Bottom-up inventories incomplete/dated

Year	Milestone (author, venue)	Contribution	Limitation it leaves
2015	Pant et al., *Atmos. Environ.*	PMF PM2.5 source apportionment, Delhi	Site-specific; heavy data requirements
2015	Monks et al., *Atmos. Chem. Phys.*	Tropospheric ozone precursor review	Confirms VOC/NOx non-linearity
2020	Tobler et al., *Sci. Total Environ.*	Organic-aerosol PMF (ACSM), Delhi	Needs online speciation instruments
2021	Xu et al., *E3S Web Conf.*	NSGA-II joint cost-emission optimisation	Optimises a model, not validated interventions
2025	Lin et al. (EGUsphere); DN-PMF India (EGU25)	RF+SHAP ozone regimes via HCHO/NO2; dispersion-normalised PMF	Still require speciation/satellite proxies
2026	**This work**	**Analytic Pareto proof: 100% NSGA-II dominance; concentration-emission fallacy**	--

---

3. Research Gaps (what the field leaves open -- and we address)

Synthesised from the open limitations in both timelines:

- **G1 -- Leakage-unaware evaluation.** Air-quality ML routinely reports random-split skill, which overstates generalisation on autocorrelated data (Just et al. 2020; Oliveira et al. 2018).
- **G2 -- Correlation read as mechanism.** Feature importance / SHAP and discovered graphs are interpreted causally without confounding or identifiability tests; SHAP itself is unstable under multicollinearity (Ye et al. 2026).
- **G3 -- Untested meteorological / boundary-layer confounding.** Causal graphs among pollutants rarely test whether edges survive meteorology (the dominant confounder).
- **G4 -- Clustering mislabelled as source apportionment.** Correlation clustering is presented as apportionment despite lacking the speciation and uncertainties that PMF requires (Paatero & Tapper 1994).
- **G5 -- Concentration-emission conflation.** "Policy optimisation" acts on ambient concentrations, not emissions, and metaheuristics are not benchmarked against exact / analytic Pareto fronts.
- **G6 -- Weak provenance QC and no open India benchmark.** Measurement-unit provenance is seldom audited, and there is no open, leakage-safe, reproducible multi-pollutant benchmark for the Indian network.

4. Our Contributions & Novelty (mapped to the gaps)

#	Contribution	Addresses	Novelty
C1	Leakage-safe, multi-protocol evaluation (random vs blocked-temporal vs leave-stations-out) with a full baseline ladder, on 204 Indian stations / 1.18M station-hours	G1	First India-scale, leakage-safe multi-pollutant benchmark
C2	Causal-discovery with bootstrap edge stability **and** an explicit meteorology-confounding test; only NO -> O3 survives	G2, G3	Couples stability + confounding testing rather than reporting a raw graph
C3	Formal distinction (and proof) that k-means correlation clustering is **not** PMF source apportionment	G4	Makes the clustering/apportionment boundary explicit and testable
C4	Analytical proof that the linear surrogate's Pareto front collapses to a 1-D segment; 100% of NSGA-II policies dominated (GD =*0.084*, HVE =*6.6*\%)	G5	First closed-form optimality audit of a heuristic AQ optimiser
C5	Formalised **concentration-emission fallacy** (negative-mass reductio)	G5	Names a common category error in data-driven AQ policy

#	Contribution	Addresses	Novelty
C6	Authoritative unit-provenance audit (ppb/µg-m-3 mix) + open reproducible benchmark + a 4-part **scientific identification gate**	G2-G6	A reusable gate for observational environmental AI

One-sentence novelty statement. To our knowledge this is the first India-scale, reproducibility-first audit that (i) proves analytically that the heuristic optimiser is dominated on the fitted surrogate, (ii) couples causal-discovery stability with meteorological-confounding tests, and (iii) formalises a scientific identification gate for observational environmental AI.

---

5. SCI / Q1 Reading List

Priority papers to read, grouped by thread. Quartiles are indicative of the journal's standing; **confirm the current quartile on Scimago (SJR) or JCR** before citing, as rankings change yearly.

Core methods (SCI / Q1)

1. Paatero & Tapper (1994), *Environmetrics* (Q1) -- PMF foundation. DOI 10.1002/env.3170050203
2. Deb et al. (2002), *IEEE Trans. Evolutionary Computation* (Q1) -- NSGA-II. DOI 10.1109/4235.996017
3. Shimizu et al. (2011), *J. Machine Learning Research* (Q1) -- DirectLiNGAM.
4. Lundberg & Lee (2017), *NeurIPS* (A*; Lundberg et al. 2020, *Nature Machine Intelligence*, Q1) -- SHAP.
5. Chen & Guestrin (2016), *KDD* (A*) -- XGBoost.

Leakage & evaluation (SCI / Q1)

6. Just et al. (2020), *Remote Sensing* (Q1) -- spatial CV vs leakage in PM2.5 ML. **Read first for G1.**
7. Roberts et al. (2017), *Ecography* (Q1) -- blocked CV for spatial/temporal data. DOI 10.1111/ecog.02881
8. Grange & Carslaw (2019), *Science of the Total Environment* (Q1) -- meteorological normalisation. DOI 10.1016/j.scitotenv.2018.10.344

Causal / explainable AQ (SCI / Q1) -- recent

9. Qi et al. (2024), *Environmental Research Letters* (Q1) -- convergent cross-mapping, PM2.5/ozone causality.
10. Wang et al. (2025), *Environmental Science and Ecotechnology* (Q1) -- XAI + causal inference, ozone rebound.
11. Ye et al. (2026), *Atmospheric Chemistry and Physics* (Q1) -- SHAP ozone attribution; multicollinearity caveat. **Read for G2.**
12. Rautela & Goyal (2024), *Scientific Reports* (Q1) -- AI/ML for air pollution management in India.
13. Chernozhukov et al. (2018), *The Econometrics Journal* (Q1) -- double/debiased ML (deconfounding).

Atmospheric chemistry & India context (SCI / Q1)

14. Monks et al. (2015), *Atmospheric Chemistry and Physics* (Q1) -- tropospheric ozone precursors.
15. Sillman (1999), *Atmospheric Environment* (Q1) -- VOC- vs NOx-limited ozone. DOI 10.1016/S1352-2310(98)00345-8
16. Pant et al. (2015), *Atmospheric Environment* (Q1) -- Delhi PM2.5 PMF apportionment.
17. Guttikunda & Calori (2013), *Atmospheric Environment* (Q1) -- Delhi emission inventory. DOI 10.1016/j.atmosenv.2012.10.040
18. Seinfeld & Pandis (2016), *Atmospheric Chemistry and Physics* (Wiley, textbook) -- reference chemistry.

How to split the reading (5 students)

- **G1 / evaluation:** #6, #7, #8
- **Causal & XAI:** #9, #10, #11, #13

- **Apportionment / PMF:** #1, #16, #17
- **Optimisation:** #2, and one applied NSGA-II emission paper (e.g. Xu et al. 2021, *E3S*)
- **Chemistry / India context:** #14, #15, #18

Each reader: note the paper's method, its evaluation protocol, and which of our gaps (G1-G6) it supports or challenges -- that mapping is exactly what strengthens our Introduction.